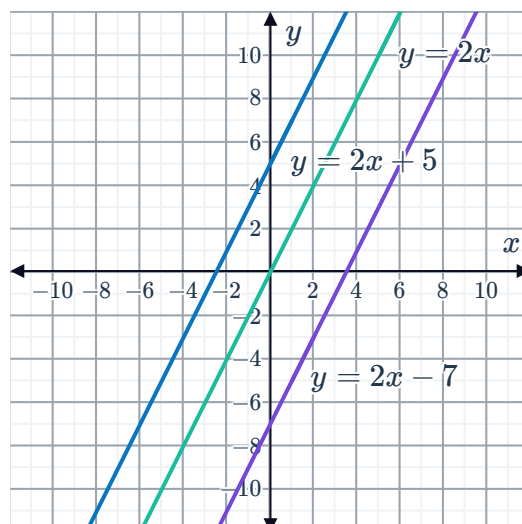


Parallel lines



- 1 The equations $y = 2x$, $y = 2x + 5$ and $y = 2x - 7$ have been graphed on the same number plane:



- a What do all of the equations have in common?
- b What do all the lines have in common?

- 2 State whether the following pairs of lines are parallel:

- | | |
|--|-----------------------------------|
| a $y = -3x - 2$ and $y = -3x + 9$ | b $y = -2x - 5$ and $y = -2x - 8$ |
| c $y = 7x + 8$ and $y = -5x + 8$ | d $y = 4x - 1$ and $y = 4x - 6$ |
| e $x = 4$ and $y = 5$ | f $y = 7x - 5$ and $y = -7x + 6$ |
| g $y = 3x - 5$ and $y = -\frac{1}{3}x + 6$ | h $y = -8x - 2$ and $9x + 7$ |

- 3 State whether the following lines are parallel to $y = 7x + 3$:

- | | | | |
|----------------|-----------------|----------------|-----------------|
| a $y = 7x - 3$ | b $y = 6x + 3$ | c $y = 7x + 4$ | d $y = 7x$ |
| e $y = 6x + 4$ | f $y = -7x + 3$ | g $y = 3x + 7$ | h $y = -3x - 7$ |

- 4 State whether the following lines are parallel to $y = -3x + 2$:

- | | | | |
|------------|-----------------|------------------|----------------|
| a $y = 3x$ | b $-3y - x = 5$ | c $y = -10 - 3x$ | d $y + 3x = 7$ |
|------------|-----------------|------------------|----------------|

- 5 State whether the following lines are parallel to $y = 9x + 2$:

- | | | | |
|------------|-----------------|-----------------|----------------|
| a $y = 9x$ | b $y = -9x + 5$ | c $y = -9x + 2$ | d $y = 9x - 2$ |
|------------|-----------------|-----------------|----------------|

- 6 Consider the lines $L_1 : 5x + 8y - 10 = 0$ and $L_2 : y = -\frac{5x}{8} + 8$.

- a Find the gradient of line L_1 .
- b Find the gradient of line L_2 .
- c Are the two lines parallel?

- 7 Consider the line $y = 2x + 2$. If every point on the line is shifted 2 units up, find the equation of the new line.
- 8 Every point on a particular line is shifted 6 units down. The resulting line has equation $y = 2x - 2$. Find the equation of the original line.
- 9 Find the equation of a line described by the following information:
- Parallel to the x -axis and passes through $(-10, 2)$.
 - Parallel to the y -axis and passes through $(-7, 2)$.
 - Parallel to the line $y = -3x - 8$ and cuts the y -axis at -4 .
 - Parallel to the line $y = 8x - 3$ and cuts the y -axis at 5 .
 - Parallel to the line $y = -2x + 9$ and passes through the point $(-3, 1)$.
- 10 If the following pairs of lines are parallel, find the value of b :
- $$\begin{aligned} 0 &= 5x - 9y + 6 \\ 0 &= bx + 3y - 2 \end{aligned}$$
 - $$\begin{aligned} y &= bx + 4 \\ 2y &= \frac{4}{5}x - 7 \end{aligned}$$
- 11 The line L_1 goes through $(3, 2)$ and $(-2, 4)$.
- Find the gradient of line L_1 .
 - Find the equation of the line that has a y -intercept of 1 and is parallel to line L_1 .
- 12 The line L_1 goes through $(-9, 10)$ and $(2, -1)$.
- Find the gradient of line L_1 .
 - Find the equation of the line that passes through $(10, 8)$ and is parallel to line L_1 .
- 13 The line L_1 passes through the point $(9, -5)$ and is parallel to the line $y = -5x + 2$.
- Find the gradient of line L_1 .
 - Find the equation of line L_1 .

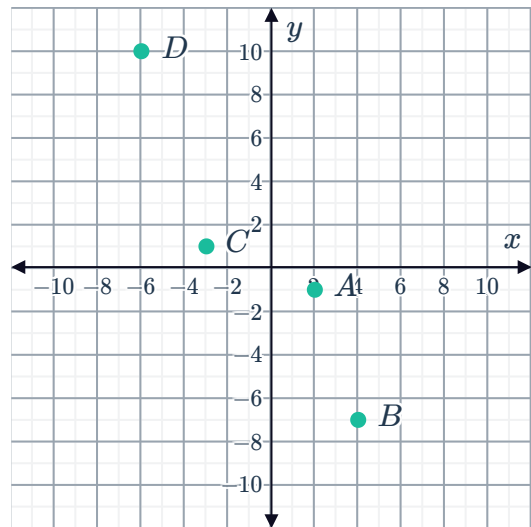


Applications

14 Consider the following points on the number plane:

- Point $A(2, -1)$
- Point $B(4, -7)$
- Point $C(-3, 1)$
- Point $D(-6, 10)$

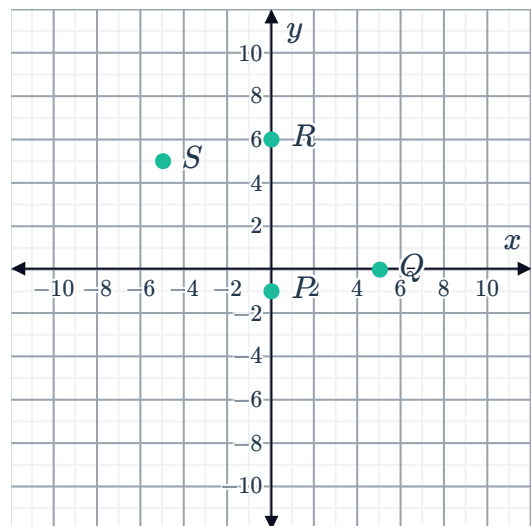
- Find the gradient of the line AB .
- Find the gradient of the line CD .
- Is the line CD parallel to AB ?



15 Consider the following points on the number plane:

- Point $P(0, -1)$
- Point $Q(5, 0)$
- Point $R(0, 6)$
- Point $S(-5, 5)$

- Find the gradient of PQ .
- Find the gradient of RS .
- Are PQ and RS parallel?
- Are QR and PS parallel?
- What type of quadrilateral is $PQRS$?



16 Determine whether the following sets of points are collinear:

- Point $A(-4, 3)$, Point $B(-2, 7)$ and Point $C(-7, -3)$.
- Point $A(-3, 1)$, Point $B(-2, 6)$ and Point $C(0, 14)$.
- Point $A(-2, 4)$, Point $B(2, 2)$ and Point $C(4, 1)$.